

# Maxim Silin

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## Cambridge Mathematics MMath | Quantitative Research | C++ & Python

University of Cambridge Mathematics graduate (BA First Class + MMath) with a strong background in **probability, statistics, physics**. With **Quantitative Research experience in FX working alongside a trading team to evaluate market opportunities** and find signals. Practical programming experience using Python and C++ to backtest strategies and analyse financial timeseries data probabilistically.

## EDUCATION

2022-2026 - University of Cambridge – Mathematics Degree

**Mathematical Tripos + Mmath: BA First Class** (Bachelors) + **Mmath Merit** (Masters).

- **Topics:** Probability, Statistics, Markov Chains, Vectors and Matrices, Linear Algebra, Differential Equations, Methods, Geometry, Variational Principles, Statistical Physics, Quantum Mechanics and others.
- **Computational Projects CATAM:** Random Binary Expansions, Monte Carlo Simulation of the Ising Model, Particle or Photon Orbits near a Black Hole, Parallel-Plate Capacitor - Laplace's Equation, and others have allowed me to delve into the **practicals of Numerical Programming**.

2020-2022 - Sixth Form - **A\* A\* A\* A\*** (four A-stars) at A level: Computer Science, Maths, Further Maths, Physics.

2015-2020 - School - **10 GCSEs, all 9's**. Gold at Maths Challenge.

## PROFESSIONAL EXPERIENCE

**07/2026 - 08/2026 - Quantitative Research Intern at Larkin Research Ltd, backed by ARB Trading Group**

Worked with an HFT Portfolio Manager to identify and **evaluate alpha opportunities in spot FX**.

- Researched **market microstructure across EBS and CBOE FX**, focusing on **arbitrage, market making, adverse selection and execution dynamics**.
- Investigated potential cross-venue trading opportunities, and whether these remained viable after considering fills and execution constraints.
- Contributed to discussions with the trading team on inventory exposure, volume clustering, jump prediction and potential ratcheting strategies. **Suggested new strategy directions**.

Inspired by trying to predict future fx market movements, **built a timeseries processing library in C++**.

- Untimed to timed timeseries conversions, resampling methods, **custom averaging** and **smoothing** methods for different signal types, with custom smoothing Kernels.
- Included an ability to **transform, combine, and find correlations between signals**. This enabled **rolling-window Fourier decomposition** of a signal - visualised as a 'rainbow' signal, travelling through time. The signal did not display stable behaviour, but itself oscillated.
- Added the ability to generate **sampled frequency distributions** from timeseries data to answer questions such as whether the previous gradient probabilistically determines the next on average.

**07/2024 - 09/2024, Software Internship – Toku Co, On-site, Singapore.**

- Worked as a Software Intern at a technology company, immersed myself in the culture.
- Programmed a large number of rules in Node.js. Worked with regex expressions and considered various approaches and benefits/drawbacks. Implemented a Python parser to aid in translating the rules, while ensuring correctness after translation.

**3-Sphere Project Computer Science A level Project**

[maxicl14.github.io/A-spherical-world-project/canvas.html](https://maxicl14.github.io/A-spherical-world-project/canvas.html)

Programmed simulation of a 4D sphere world. Implementing mathematical calculations in a C++ like language (WebGL) to make a game where a rocket is traversing a 4-dimensional sphere on the sphere's surface (3D warped), seeing planets, and bumping into them (collision detection & prevention). Python program to generate planet vertices.

### Python AES Encryption Research Project

Used Python to program AES128 encryption algorithm. Used to encrypt a picture of my friend's face with AES128. Noticed patterns of regularity in the encrypted picture and realised that this emerged as a result of white patches of light in the original picture. Realised the importance of larger block sizes and preprocessing data to prevent exploits.

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### KEY SKILLS

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**Mathematics:** Probability & Statistics, Markov Chains, Stochastic Calculus, Mathematical Modelling, Monte Carlo, Differential Equations & Methods, Geometry.

**Quantitative Research:** FX Market Microstructure, Signal Processing, Statistics, Backtesting, Time-Series Analysis.

**Programming:** Python, NumPy, Pandas, Scikit-learn, C++, MATLAB

**Others:** Javascript, AWS, Node.js, React.js, REST APIs, MongoDB, Linux, Git, Github, SQL, OOP, Excel, LibreOffice.

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